higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

BUILDING AND STRUCTURAL SURVEYING N4

(8060034)

2 December 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages and 1 formula sheet.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
BUILDING AND STRUCTURAL SURVEYING N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

Define each of the following terms:

1.1 Pentaprism

1.2 Direction

1.3 Chain surveying

1.4 Coordinates

1.5 Prime meridian

(5 × 2) **[10]**

QUESTION 2

2.1 Name FOUR methods of measurement. (4)

2.2 Name THREE areas into which a surveyor's functions are divided. (3)

2.3 State FIVE points to observe when taking field notes. (5)

[12]

QUESTION 3

3.1 The formula for tension or pull is as follows:

$$\text{Cp} = 1\text{m}\left(\frac{P - PO}{AE}\right)$$

What does each letter in the formula stand for? (5)

3.2 A triangular-shaped farm is 1,560 ha. The length of BC is 250 m and angle BCA is 30°.

Calculate the length of side AC. (5)

3.3 What is the size of a 45° angle in centesimal? (4)

3.4 The dimensions of a rectangular plot are 160 m × 80 m.

Calculate the area of the plot. (4)

[18]

QUESTION 4

4.1 Explain how to use an Abney level and two ranging rods to measure ground slope. (5)

4.2 Briefly explain how an electronic distance-measuring instrument works. (3)

4.3 State SEVEN points regarding the care of steel tapes. (7)

[15]

QUESTION 5

5.1 It is required to set off an exact distance to 100 m with a 50 m steel tape known to be 0,03 m too long.

Calculate the distance to be measured off with this incorrect tape to get an exact length of 100 m. (4)

5.2 A distance of 600 m was measured with a steel tape at a temperature of 15 °C.

Calculate the correct length if the standard temperature is 20 °C and the coefficient of linear expansion is $1{,}13 \times 10^{-5}$/°C. (4)

5.3 A length of 95,45 m was measured between two points at an angle of 7° 40' 20" from the horizontal.

Calculate the horizontal distance between the points. (4)

5.4 A length of 300 m was measured with a steel tape at an altitude of 1 500 m below sea level. The radius of the earth is 6 373 km.

Calculate the correct horizontal distance. (4)

[16]

QUESTION 6

6.1 Give FIVE examples of levelling equipment. (5)

6.2 What is a parallax and how can it be removed? (7)

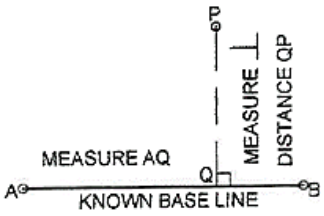
[12]

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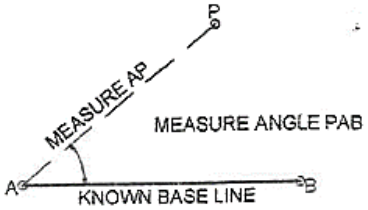
QUESTION 7

7.1 Name each of the following methods of fixing a point:

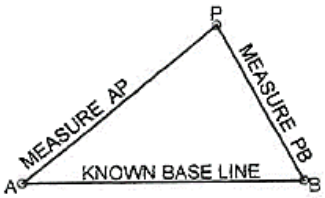
7.1.1



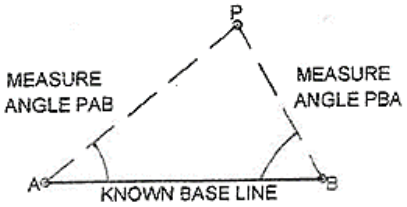
7.1.2



7.1.3

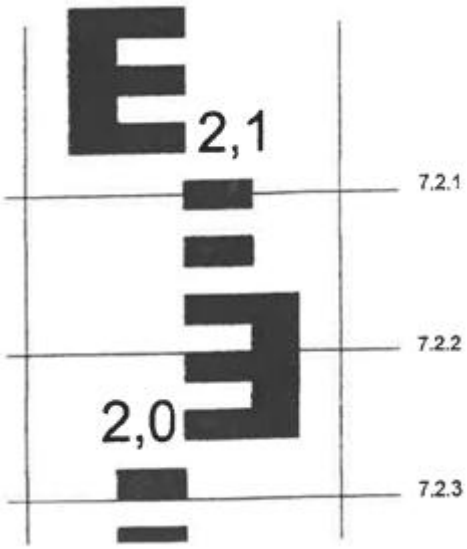


7.1.4



(4 × 1) (4)

7.2 Part of a levelling staff is shown below.



Give the readings on the levelling staff by writing only the answer next to the question number (7.2.1–7.2.3) in the ANSWER BOOK. (3)

[7]

QUESTION 8

Show, by means of simple sketches, how each of the following details are shown on a detailed map:

8.1 Bush or scrub

8.2 Bench marks 

8.3 Stream boundaries

8.4 Fence on cadastral boundaries

8.5 Hedges

(5 × 2) **[10]**

TOTAL: 100

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FORMULA SHEET

Any other applicable formula may also be used.

1. $A = {}^{1}/_{2}\, b \times \perp h$

2. $A = \sqrt{s(s-a)(s-b)(s-c)}\,,\ \frac{a+b=c}{2}$

3. $A = \frac{(a+b)\times h}{2}$

4. $A = a \times h$

5. $A = \frac{d}{3}[(y_1 + y_n) + 2(y_3 + y_5 + \ldots + (y_n - 2) + 4(y_2 + y_4 + \ldots + y_n - 1)]$

6. $A = d\,[(\frac{y_2 + y_n}{2} + y_2 + y_3 + \ldots + y_{n-1}]$

7. $C = L \times e \times (t_m - t_s)$

8. $C = \frac{W^2 L^3}{24T^2}$

9. $C = L\,(1 - \cos\theta)$

10. $C = H\,(\sec - 1)$

11. $C = \frac{LH}{R}$

12. $A = \frac{1}{2}ab\sin C$

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